Electric filed

When an electric charge is placed at some point in the space, the surrounding place of the charge will be affected. The place around the charge within which its effect exists is called the ***electric filed****.*

The ***intensity of the electric field*** is defined as, '*the force experienced by a unit positive (for convenience) charge placed at the point at which the field is to be determined.'*

If a positive test charge q_0 is placed at a point in the electric field, then the electric field intensity $\vec{E}$.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

Where, $\vec{F}$ is the force experienced by the charge q_0. The direction of the field is along the direction of the force. The SI unit of $\vec{E}$ is Newton/Coulomb (NC^{-1}).

Lines of force: Electric lines of force is defined as a way or path, which may be straight or curved, so that the tangent at any point to it gives the direction of the electric field intensity at that point.

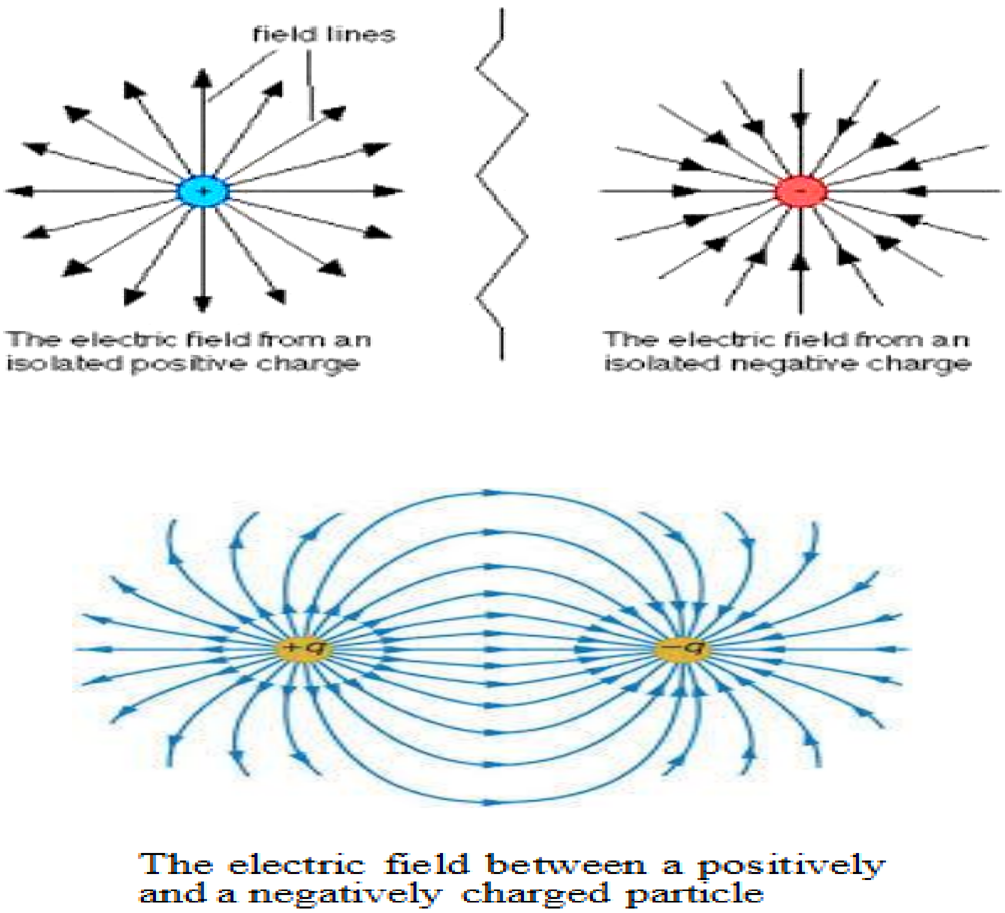


Fig. 1

Calculation of electric field

Electric field due to a point charge:

Let a test charge q_0 is placed at a distance r from the point charge q. The magnitude of the force acting on q_0 is given by the Coulomb's law, *i.e.*,

$$F = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2} \qquad (1)$$

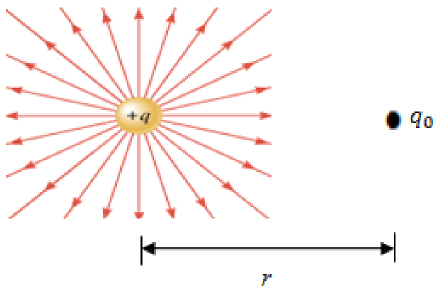


Fig. 2

Then the electric field intensity E is

$$E = \frac{F}{q_0}$$

$$\therefore \qquad E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \qquad (2)$$

This is the expression for the electric field intensity due to a point charge.

We can find the *net*, or *resultant*, electric field due to more than one point charge. If we place a positive test charge q_0 near n point charges $q_1,\ q_2, \ldots \ldots \ldots . q_n$ then the net electric field at the position of the test charge is

$$E = E_1 + E_2 + \cdots\cdots\cdots + E_n = \frac{F_1}{q_0} + \frac{F_2}{q_0} + \cdots\cdots\cdots + \frac{F_n}{q_0}$$

Electric filed due to a long straight uniform charged wire

Let us consider a long wire having charge λ coulombs per meter oriented as shown in fig.3. The electric filed at P due to the charge $dq = \lambda dx$ is

$$dE = \frac{\lambda dx}{4\pi\varepsilon_0 r^2} \qquad (1)$$

If the wire is infinitely long and be symmetrically placed on both right and left of the perpendicular PO, then for any element of charge $dq = \lambda dx$ to the right of O there is an equal element in a symmetrical position to the left. Consequently, the resultant field at P can only be directed outward along the perpendicular OP. Thus only the component $dE\cos\theta$ exists.

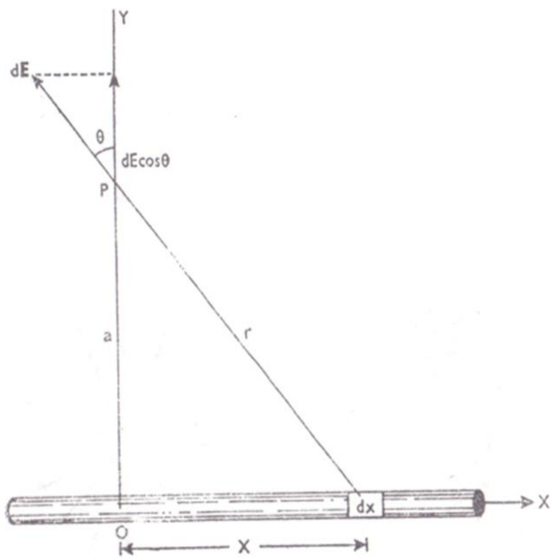


Fig. 3.

$$dE\cos\theta = \frac{\lambda}{4\pi\varepsilon_0}\frac{dx}{r^2}\cos\theta$$

$$\therefore \qquad dE\cos\theta = \frac{\lambda}{4\pi\varepsilon_0}\frac{dx}{r^2}\frac{a}{r} = \frac{\lambda a dx}{4\pi\varepsilon_0 r^3} \qquad (2)$$

$$Where, \cos\theta = \frac{a}{r}$$

It can contribute to the resultant field at P. Therefore, the electric field at P is

$$E_P = \int_{-\infty}^{\infty} dE \cos\theta$$

$$= \int_{-\infty}^{\infty} \frac{\lambda a dx}{4\pi\varepsilon_0 r^3}$$

$$= \int_{-\infty}^{\infty} \frac{\lambda a dx}{4\pi\varepsilon_0 (a^2 + x^2)^{3/2}}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\lambda a^2 \sec^2\theta \, d\theta}{4\pi\varepsilon_0 (a^2 + a^2\tan^2\theta)^{3/2}}$$

Since, $tan\,\theta = \frac{x}{a}$

$$x = a\,tan\,\theta$$

$$dx = a\sec^2\theta\, d\theta$$

When $x = \infty, then\ \theta = \frac{\pi}{2}$

When $x = -\infty, then\ \theta = -\frac{\pi}{2}$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\lambda a^2 \sec^2\theta\, d\theta}{4\pi\varepsilon_0 a^3 \sec^3\theta}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\lambda\cos\theta\, d\theta}{4\pi\varepsilon_0 a}$$

$$= \frac{\lambda}{4\pi\varepsilon_0 a}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\theta\, d\theta$$

$$= \frac{\lambda}{4\pi\varepsilon_0 a}[\sin\theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \frac{\lambda}{4\pi\varepsilon_0 a}(1+1)$$

$$\therefore \qquad E_P = \frac{\lambda}{2\pi\varepsilon_0 a} \qquad (3)$$

From equation (3) it is evident that in the case of long straight wire, the electric field is inversely proportional to the first power of the distance from the wire and is not inversely proportional to the square of the distance as in the case of a point charge.

Electric filed at a point on the axis of a charged circular turn of wire

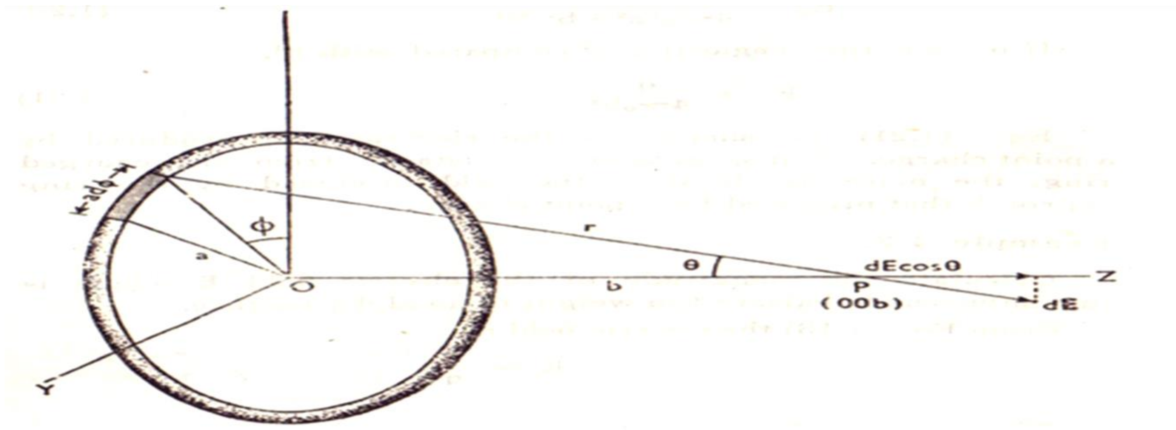


Fig. 4. Electric filed at a point on the axis of a charged circular turn of wire

Let us consider a circular turn of wire of radius a charged with λ coulombs per meter. The electric filed at $P(0,0,b)$ due to the element of charge $dq = \lambda(ad\varphi)$ is

$$dE = \frac{\lambda a d\varphi}{4\pi\varepsilon_0 r^2} \qquad (1)$$

The component $dE\cos\theta$ contribute only because of symmetry.

Therefore

$$dE\cos\theta = \frac{\lambda a d\varphi}{4\pi\varepsilon_0 r^2}\cos\theta = \frac{\lambda ab}{4\pi\varepsilon_0 r^3}d\varphi \qquad (2) \qquad Where, \cos\theta = \frac{b}{r}$$

Therefore, the electric field at P due to the charge q is

$$E_P = \int_0^{2\pi} \frac{\lambda ab}{4\pi\varepsilon_0 r^3} d\varphi$$

$$E_P = \frac{\lambda ab}{4\pi\varepsilon_0 r^3}\int_0^{2\pi} d\varphi$$

$$E_P = \frac{\lambda ab}{4\pi\varepsilon_0 r^3}[\varphi]_0^{2\pi}$$

$$E_P = \frac{\lambda ab}{4\pi\varepsilon_0 r^3}(2\pi - 0)$$

$$E_P = \frac{\lambda ab}{2\varepsilon_0 r^3}$$

$$\therefore \qquad E_P = \frac{\lambda ab}{2\varepsilon_0 (a^2+b^2)^{3/2}} \qquad (3) \qquad Where, r = \sqrt{a^2+b^2}$$

If the total charge on the ring is $q = 2\pi\lambda a$, then

$$E_P = \frac{qb}{4\pi\varepsilon_0 (a^2 + b^2)^{3/2}} \qquad (4)$$

If $b \gg a$, then neglecting a^2compared with b^2

$$E_P = \frac{qb}{4\pi\varepsilon_0 b^3}$$

$$\therefore \qquad E_P = \frac{q}{4\pi\varepsilon_0 b^2} \qquad (5)$$

From equation (5) it is evident that in the case of circular turn of wire, the electric field is inversely proportional to the square of the distance from the centre of the ring. Thus, greater the distance from the charged ring, the more nearly does the field produced by the ring approach that produced by a point charge.

Mathematical problems

Problem-1: *What is the magnitude of the electric field strength E such that an electron, placed in the field would experience an electrical force equal to its weight. (Mass of the electron=* $9.1 \times 10^{-31} kg$*)*

Solution: We know,

$$E = \frac{F}{q_0} = \frac{mg}{e}$$

$$Or \qquad E = \frac{9.1 \times 10^{-31} \times 9.8}{1.6 \times 10^{-19}}$$

$$\therefore \qquad E = 5.57 \times 10^{-11}\ N/C\ (\boldsymbol{Ans})$$

Here,

$m = 9.1 \times 10^{-31} kg$

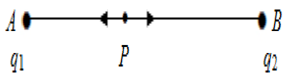
$g = 9.8\ ms^{-2}$

$q_0 = e = 1.6 \times 10^{-19}\ C$

$E = ?$

Problem-2: *Two point charges of unknown magnitude and sign are at a certain distance apart. The electric field strength is zero at one point on the line joining them. What can you conclude about the charge?*

Solution: As the electric field is zero at a point *between* the two charges, either both the charges are positive or both of them are negative.



When both the charges say q_1 and q_2 are positive and placed at points *A* and *B* and *P* is the point between them where electric field is zero, then the electric filed at *P* due to the charge $+q_1$ at *A*

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{AP^2} \qquad along\ AP\ is\ produced$$

and electric filed at *P* due to the charge $+q_2$ at *B*

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{BP^2} \qquad along\ BP\ is\ produced$$

i.e., E_1 *and* E_2 act in opposite directions. They will cancel each other and the resultant field will be zero when

$$\frac{1}{4\pi\varepsilon_0}\frac{q_1}{AP^2} = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{BP^2}$$

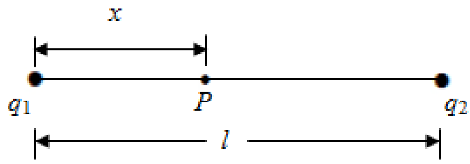
$$\therefore \qquad \frac{q_1}{q_2} = \frac{AP^2}{BP^2}$$

Similar treatment holds when both the charges are negative.

But if one charge say q_1 is positive and the other charge q_2 is negative, then E_1 *and* E_2 will both act in same direction *i.e.*, *AP* produced and their resultant will never be zero.

Problem-3: *Two charges* $q_1 = 1.0 \times 10^{-6}$ *coul and* $q_2 = 2.0 \times 10^{-6}$ *coul are kept 10 cm away from each other. At what point the line joining the two charges is the electric field strength zero?*

Solution: The point (x) must lie between the charges because only here do the forces exerted by q_1 and q_2 on a test charge oppose each other. If E_1 is the electric field strength due to q_1 and E_2 is the electric filed strength due to q_2, we must have



$$E_1 = E_2$$

$$\therefore \qquad \frac{1}{4\pi\varepsilon_0}\frac{q_1}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{(l-x)^2}$$

$$or \qquad \frac{q_1}{x^2} = \frac{q_2}{(l-x)^2}$$

$$or \qquad \frac{(l-x)^2}{x^2} = \frac{q_2}{q_1}$$

$$or \qquad \frac{l-x}{x} = \sqrt{\frac{q_2}{q_1}}$$

$$or \qquad \frac{l}{x} - 1 = \sqrt{\frac{q_2}{q_1}}$$

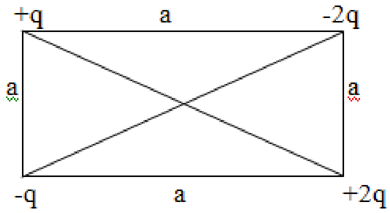
$$or \qquad \frac{l}{x} = 1 + \sqrt{\frac{q_2}{q_1}}$$

$$or \qquad x = \frac{l}{1+\sqrt{\frac{q_2}{q_1}}} = \frac{10}{1+\sqrt{\frac{2.0\times10^{-6}}{1.0\times10^{-6}}}}$$

$$\therefore \qquad x = 4.1\ cm$$

At 4.1 cm from the charge q_1 the electric field strength zero. ***(Ans)***

Problem-4: *What is the electric field E in magnitude at the centre of the following square? Assume that* $q = 1.0 \times 10^{-8} coul$ *and* $a = 5.0\ cm$.



Solution: The distance of each charge from the centre of the square is $\frac{a}{\sqrt{2}}$ or 3.53 cm.

The electric field E at the centre of the square is

$$E = \frac{1}{4\pi\varepsilon_0}\sum\frac{q}{r^2}$$

$$Or \qquad E = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r^2}+\frac{q_2}{r^2}+\frac{q_3}{r^2}+\frac{q_4}{r^2}\right)$$

$$Or \qquad E = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2}-\frac{2q}{r^2}+\frac{2q}{r^2}-\frac{q}{r^2}\right)$$

$$Or \qquad E = \frac{1}{4\pi\varepsilon_0}\times 0$$

$$\therefore \qquad E = 0$$

Thus the electric field E at the centre of the square is zero. ***(Ans)***